Physicist / Engineer physicist BSc, year III / semester 1
"Optics problem solving"

MIDTERM TEST 2

04-12-2025

1 Analysis of a shearing interferometer with two-beam interference

You are examining the plane wave-shaped beam of a HeNe laser of λ_0 = 633 nm wavelength by a shearing interferometer at $\theta = 45°$ angle of incidence, in air (n = 1.0), see Fig. 1. The thickness of the plane-parallel glass plate is d = 1.0 mm, its refractive index is n' = 1.515 (Schott N-BK7). The question is how much we shall change the angle of incidence ($\Delta\theta$), i.e. tilt the glass plate, so that the phase difference between the two interfering beams changes exactly by $\Delta\delta = 2\pi$. What is the angle of refraction θ'? How can we express the optical path difference (*OPD*) between the two waves along the D-D' wavefront as a function of $\cos\theta'$? What is the phase difference between the two waves (δ)? Since the tilt angle is to be small, express the $\Delta\delta$ change as a function of $\Delta\theta'$ in first-order approximation! Based on this what $\Delta\theta'$ corresponds to a phase difference change of 2π? By Snell's law what is $\Delta\theta$ in [°]? **30 p**

$$\sin\theta = n'\sin\theta' \tag{1}$$

$$\theta' = \operatorname{asin}\frac{\sin\theta}{n'} = 27.82° \tag{2}$$

$$OPD = 2\frac{d}{\cos\theta'}n' - 2d\tan\theta'\sin\theta \tag{3}$$

$$OPD = 2\frac{d}{\cos\theta'}n' - 2d\tan\theta'\sin\theta'\,n' \tag{4}$$

$$OPD = 2dn'\left(\frac{1}{\cos\theta'} - \tan\theta'\sin\theta'\right) = 2dn'\cos\theta' \tag{5}$$

Due to Fresnel reflection:

$$\delta = OPD\frac{2\pi}{\lambda_0} - \pi = \frac{2\pi}{\lambda_0}2dn'\cos\theta' - \pi \tag{6}$$

$$\Delta\delta = \frac{\mathrm{d}\delta}{\mathrm{d}\theta'}\Delta\theta' = -\frac{2\pi}{\lambda_0}2dn'\sin\theta'\,\Delta\theta' = 2\pi \tag{7}$$

$$\Delta\theta' = -\frac{\lambda_0}{2dn'\sin\theta'} \tag{8}$$

$$\Delta\theta' = -\frac{0{,}000633\ \text{mm}}{2\cdot 1\ \text{mm}\cdot 1.515\cdot\sin 27.82°} = -0.4476\ \text{mrad} = -0.02565° \tag{9}$$

$$\sin(\theta + \Delta\theta) = n'\sin(\theta' + \Delta\theta') \tag{10}$$

$$\Delta\theta = \operatorname{asin}(n'\sin(\theta' + \Delta\theta')) - \theta = -0.04859°. \tag{11}$$

2 Examination of the spectrum of a superluminescent LED by a diffraction grating

You are analyzing the power spectrum of the $\lambda_0 = 830$ nm wavelength light of a superluminescent LED by a $D = 10$ mm aperture-size transmissive diffraction grating in air. The LED illuminates the grating as a plane wave arriving at $\theta = 45°$ angle of incidence (see Fig. 2). In the measurement we want to have the $m = -1$ diffraction order exit the grating in a direction normal to it. In this case how much is θ'? What is the optical path difference (OPD) between two neighboring grating elements along the B-C wavefront being normal to the θ' direction (lower – upper)? From this how can we express the phase difference, and what is the condition for constructive interference? What is the grating constant (a) necessary for the perpendicularly exiting order, and the groove density (f_y)? What is the grating resolution ($\mathcal{R}$)? **25 p**

$$OPD = n'a\sin\theta' - na\sin\theta \quad ; \quad n = n' = 1 \tag{12}$$

$$\delta = \frac{2\pi}{\lambda_0}a\sin\theta' - \frac{2\pi}{\lambda_0}a\sin\theta = m2\pi \quad ; \quad m = -1 \tag{13}$$

$$\frac{a}{\lambda_0}\sin\theta' - \frac{a}{\lambda_0}\sin\theta = -1 \tag{14}$$

$$a = \frac{\lambda_0}{\sin\theta - \sin\theta'} \quad ; \quad \theta = 45° \quad ; \quad \theta' = 0° \tag{15}$$

$$a = \frac{\lambda_0}{\sqrt{2}/2 \ - 0} = \frac{0.830\ \mu\text{m}}{\sqrt{2}}2 = 1.174\ \mu\text{m} \tag{16}$$

$$f_y = \frac{1}{a} = \frac{1}{0.001174\ \text{mm}} = 852\frac{\text{vp}}{\text{mm}} \tag{17}$$

$$\mathcal{R} \triangleq \frac{\lambda_0}{\Delta\lambda_0} = |mN| = \left|m\frac{D}{a}\right| = \left|-1\frac{10\ \text{mm}}{0.001174\ \text{mm}}\right| = 8519. \tag{18}$$

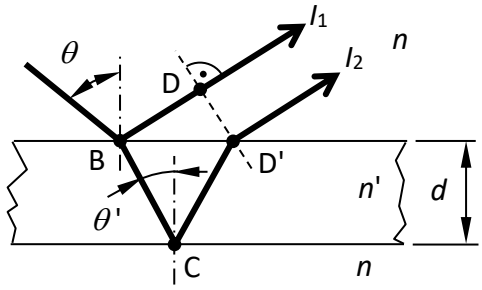


Figure 1

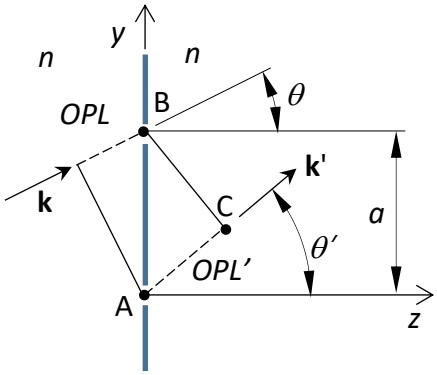


Figure 2

3 Determination of the theoretical resolution of the Vera C. Rubin observatory

The Vera C. Rubin observatory was put into operation in 2025 containing the world's largest-resolution telescope (its image sensor is of 3200 Mpixel). This property is due to its large field-of-view of Ø3.5° and the small focal spot resulting from the huge aperture. Calculate the Airy-radius value (θ_{Airy}) corresponding to the D = 8.4 m diameter of the main mirror in arc seconds, at $\lambda = 1.05\ \mu\text{m}$ wavelength! If we define the resolution ($\mathcal{R}$) as the ratio of the field-of-view and the θ_{Airy} Airy radius, then what is the theoretical maximum of the resolution of the instrument? (Theoretical it is, since in reality the resolution is one order of magnitude worse than this due to geometrical aberrations and atmospheric turbulence.) **20 p**

$$R_{\text{Airy}} = 1.22\frac{\lambda z'}{D} \tag{19}$$

$$\tan\theta_{\text{Airy}} = 1.22\frac{\lambda}{D} = 1.22\frac{0.00105\ \text{mm}}{8400\ \text{mm}} = 1.525\cdot 10^{-7} \tag{20}$$

$$\theta_{\text{Airy}} = 8.738\cdot 10^{-6°} = 0.03145" \tag{21}$$

$$\mathcal{R} = \frac{3.5°}{8.738\cdot 10^{-6°}} = 400567. \tag{22}$$

4 Investigation of the far-field diffraction of an inhomogeneously lit aperture

You are illuminating a one-dimensional Σ aperture with the below field distribution:

$$E_1(x) \coloneqq \begin{cases} E_0 \cdot e^{-ax} & \text{if } x \geq 0, \quad x \to +\infty \\ 0 & \text{if } x < 0, \quad x \to -\infty \end{cases}$$

Determine the spatial dependence of the complex diffraction field distribution ($E_1(x')$) taking place on a screen being at a large distance z'. Express the integral value in real + imaginary part form! According to this, without using integration, what is the ($E_2(x')$) spatial distribution in the far field in case of the *reversed* field distribution?

$$E_2(x) \coloneqq \begin{cases} 0 & \text{if } x > 0, \quad x \to +\infty \\ E_0 \cdot e^{+ax} & \text{if } x \leq 0, \quad x \to -\infty \end{cases}$$

What is the resultant field ($E(x')$) of these two distributions? **25 p**

$$E_1(x') = \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}x'^2}}{\sqrt{i\lambda z'}} \int_{\Sigma} E_1(x) \cdot e^{-i\frac{k}{z'}x'x} \mathrm{d}x \tag{23}$$

$$\int_{\Sigma} E_1(x) \cdot e^{-i\frac{k}{z'}x'x} \mathrm{d}x = E_0 \int_0^{\infty} e^{-ax} e^{-i\frac{k}{z'}x'x} \mathrm{d}x \tag{24}$$

$$\int_0^{\infty} e^{-ax} e^{-i\frac{k}{z'}x'x} \mathrm{d}x = -\frac{1}{a + ikx'/z'} \left[e^{-ax} e^{-i\frac{k}{z'}x'x} \right]_0^{\infty} = \frac{1}{a + ikx'/z'} = \tag{25}$$

$$= \frac{a - ikx'/z'}{a^2 + (kx'/z')^2} \tag{26}$$

$$E_1(x') = E_0 \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}x'^2}}{\sqrt{i\lambda z'}} \left(\frac{a - ikx'/z'}{a^2 + (kx'/z')^2} \right) \tag{27}$$

$$E_2(x') = E_0 \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}x'^2}}{\sqrt{i\lambda z'}} \left(\frac{a + ikx'/z'}{a^2 + (kx'/z')^2} \right). \tag{28}$$

$$E(x') = E_1(x') + E_2(x') = E_0 \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}x'^2}}{\sqrt{i\lambda z'}} \left(\frac{2a}{a^2 + (kx'/z')^2} \right). \tag{29}$$